

collected about them — for example, combining users' product search with their purchase history and budget range. No matter how complex data is, it needs to be combined and analysed properly for the best decision making.

ELK is an acronym for three open source projects. The first is Elasticsearch, which is the distributed search and analytics engine. Then comes Logstash, which is the data pipeline that helps ingest

data from multiple sources, transforms it and sends it to data stashes like Elasticsearch. And finally, Kibana lets users visualise data stored in Elasticsearch for better understanding and business interactions. ELK is easy to set up, scalable and efficient, and provides near real-time search analytics. Its capabilities include data feature enhancing, visual analytics and alerting mechanisms with the help of Logstash and Kibana.

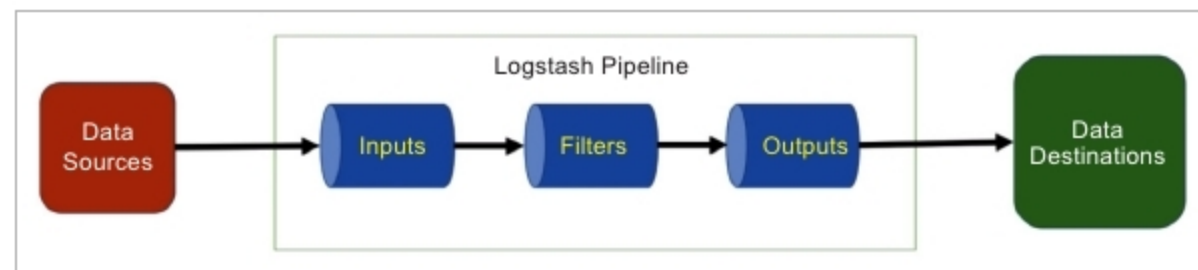


Figure 1: Logstash pipeline

```

176.111.226.153 - - [17/Apr/2019:23:59:15 +0000] "GET /item/software/1891 HTTP/1.1" 200 85
"http://www.google.com/search?ie=UTF-8&q=google&client=psy-ab&q=Software+Software&q=Software+Software&q=f&aqi=g-vl1&aql=8pbx=1&bav=on.2,or.r_gc.r_pw.r_qf.,cf.osb&biw=4849&bih=260" "Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/535.11 (KHTML, like Gecko) Chrome/17.0.963.56 Safari/535.11"
  
```

Figure 2: Example server log

```

input {
  file {
    path => "/Users/alakazam/Documents/projects/elk/data/apache-daily-access.log"
    type => "logs"
    start_position => "beginning"
  }
}
filter {
  grok {
    match => {
      "message" => "%{COMBINEDAPACHELOG}"
    }
  }
  mutate {
    convert => { "bytes" => "integer" }
  }
  date {
    match => [ "timestamp", "dd/MMM/YYYY:HH:mm:ss Z" ]
    locale => en
    remove_field => "timestamp"
  }
  geoip {
    source => "clientip"
  }
  useragent {
    source => "agent"
    target => "useragent"
  }
}
output {
  elasticsearch {
  }
  stdout {
    codec => dots
  }
}
  
```

Figure 3: Configuration file for Logstash

A hands-on guide on the ELK stack can be found at github.com/Alakazam03/ELK-Tutorial.

1. Logstash

Logstash is a server-side data processing pipeline used to ingest, parse and then transform data, before outputting it to Elasticsearch. It provides various filters that enhance the insights and human readability.

Logstash ingests the structured and unstructured data from various sources such as logs, system metrics and Web applications, by using the input plugins available. It transforms and combines the data using many of the available filters to a common format for more powerful analysis and business value.

Filter examples

- *grok*: Derives structure from unstructured data
- *geo-ip*: Extracts information from IP addresses
- *date*: Extracts date and time from standard

Logstash structures and enhances the data collected from different sources to output it into Elasticsearch, which is a NoSQL database that can store data and provide real-time search over that multi-valued data. Logstash has a variety of outputs that let you route transformed data where you want it, flexibly.

2. Elasticsearch

Elasticsearch is a distributed open source search engine released under the Apache licence. It is a REST API layer over Apache's Lucene (<https://lucene.apache.org/>). It provides horizontal scalability, reliability and real-time search through the documents. Elasticsearch is able to search faster because it uses indexing to go over documents. Lucene (the storage engine of Elasticsearch) stores the data using a technique called inverted indexing. This storage architecture lets Elasticsearch provide functionalities like aggregation (similar to GROUPBY in SQL), which helps